SPECIFICATIONS OF SFD64KX32M64P4

Rev 010A

Date : 02/15/2006

Writer : Shang-Hsuan Liu

TSMC

SFD64KX32M64P4

64k x 32 Embedded Flash Memory

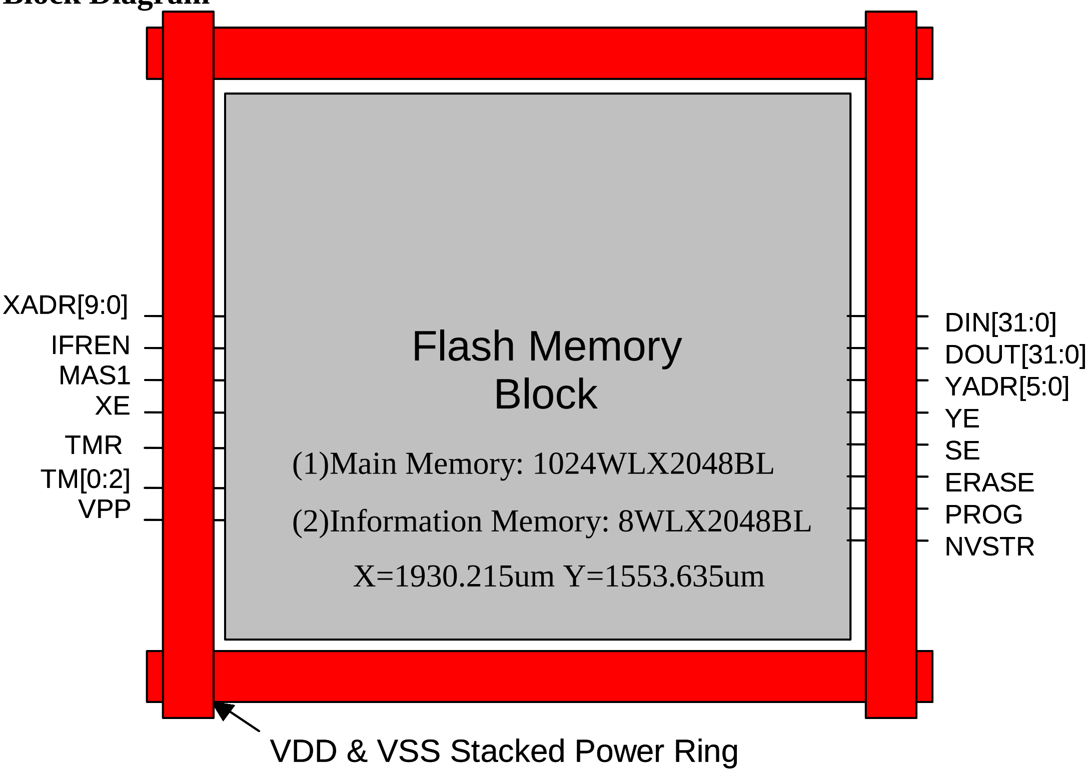
General Description

The SFD64KX32M64P4 is a CMOS page erase, double word (32-bit) program Embedded Flash Memory which is partitioned into two memory blocks. One is the main memory block, the other is the information block .The main memory block is organized as 65,536-word by 32-bit. The information block is organized as 512-word by 32-bit. The information block can be used for the storage of device information. The page erase operation erases all double words within a page. A page is composed of four adjacent rows for main memory block and information block .The split gate cell design and thick oxide tunneling injector attain better reliability and manufacturability compared with alternative approaches. The SFD64KX32M64P4 erases and programs with a 1.8-volt only power supply.

Features

- Single 1.8-volt power supply
- Endurance: 1000 cycles (min) with 20ms erase and 20us program time at 1.8Vcc
- Memory Organization 64kx32 + 512x32
- Page Erase Capability: 1024 bytes per page
- Access Time : 35ns(max)
- Current
  Operating : 21mA(max)
  Standby : 50uA(max) (Tj=-40C~125C)
  15uA(max) (Tj=-40C~85C)
- Greater than 100 years Data Retention under Room Temperature
- Fast Page Erase/Word Program Operation
- Word Program Time: 20us(min)
  Page Erase Time: 20 ms(min)
  Mass Erase Time: 20 ms(min)

Block Diagram



Connector Description

User Mode[1]

Connector name	Direction	Description
XADR[9:0]	I	X address input, access rows, XADR[1:0] select one row within a page of main memory block or information block.
YADR[5:0]	I	Y address input, access a word within a row.
DIN[31:0]	I	Data input bus.
DOUT[31:0]	O	Data output bus.
XE	I	X address enable, all rows are disabled when XE=0.
YE	I	Y address enable, YMUX is disabled when YE=0.
SE	I	Sense amplifier enable.
IFREN[3]	I	Information block enable
ERASE	I	Defines erase cycle.
MAS1	I	Defines mass erase cycle, erase whole block.
PROG	I	Defines program cycle.
NVSTR	I	Defines non-volatile store cycle.
VDD[2]	I	Power supply
VSS[2]	I	Ground

Notes:
1. Need to multiplex all control, address and data signals to package pins to facilitate direct memory test.
2. If routing power width is as wide as 30um, connecting only one set of VDD/VSS is allowable. If not, the total width of power line connecting to two sets or three sets of VDD/VSS should not be less than 30um .
3. When access the information block, the XADR[1:0] select one row within a page and XADR[3:9] are don’t care.

Test Mode[1]

Connector name	Direction	Description
TMR[2]	I	Register reset for test mode. In user mode, it should be connected to VDD.
VPP	I/O	High voltage pin used in test mode and operating in 0~13V. A high voltage pad with a I/O path for VPP pin will be provided by TSMC.
TM[1:0]	I/O	Analog pins used in test mode and operating in 0~VDD. These analog pins are high impedance in user mode. In user mode, being pulled to GND or floating is strongly recommended.
TM[2][3]	I/O	Analog pins used in test mode and operating in 0~3V.The analog pin with operation voltage > 3V was recommended to use for this pin. In user mode, being pulled to GND or floating is strongly recommended.

Notes:
1. Need to multiplex the test mode related signals to package pins for memory testing.
2. TMR is a digital pin and operating in 0~VDD.
3. TM[2] is used for debugging and will not be used in testing flow and user mode.

Recommended DC Operating Conditions

(T_J = -40°C to 125 °C)

Parameter	Symbol	Min.	Typ.	Max.	Unit
Supply Voltage	V_{DD}	1.62	1.8	1.98	V
Ground	V_{SS}	0	0	0	V

DC Electrical Characteristics (T_J = -40 °C to 125 °C, V_{DD} = 1.62 V to 1.98 V, V_{SS} = 0V)

Parameter	Symbol	SFD64KX32M64P4		Unit	Condition
		Min.	Max.		
Operating Current Read Program Erase Mass erase	I_{DD1}	 - - - -	 21 7 7 7	mA	Min.Cycle,Duty= 100% V_{IN} = V_{DD} or V_{SS} $I_{I/O}$ = 0mA
Static Read Current	I_{DD2}	-	8	mA	XE=YE=SE =VDD $I_{I/O}$ = 0mA
Standby Power Supply Current	I_{SB}	-	50 [note1] 15 [note2]	uA uA	XE, YE, SE, PROG, ERASE, and MAS1 are all V_{SS}. Other V_{IN} = V_{DD} or V_{SS}

Note1: T_J = -40 °C to 125 °C

Note2: T_J = -40 °C to 85 °C

Truth Table

Mode	XE	YE	SE	PROG	ERASE	MAS1	NVSTR
Standby	L	L	L	L	L	L	L
Read	H	H	H	L	L	L	L
Double Word Program	H	H	L	H	L	L	H
Page Erase	H	L	L	L	H	L	H
Mass Erase	H	L	L	L	H	H	H

IFREN Truth Table

Mode	IFREN=1	IFREN=0
Read	Read information block	Read main memory block
Double Word Program	Program information block	Program main memory block
Page erase	Erase information block	Erase main memory block
Mass erase	Erase both block	Erase main memory block

Test Mode Decoder Truth Table

Mode	CODE				
	ERASE	YE	XE	IFREN	MAS1
Test mode 1	**L**	**H**	**L**	**L**	**H**
Test mode 2	**L**	**H**	**L**	**H**	**L**
Test mode 3	**H**	**H**	**H**	**L**	**L**
Test mode 4	**H**	**H**	**H**	**L**	**H**
Test mode 5	**H**	**H**	**H**	**H**	**L**

Capacitance

Parameter	Symbol	Min.	Max.	Unit
Address & Control Input pin Capacitanc	C_{IN}	-	0.3	pF
Data Input pin Capacitance	C_{DIN}	-	0.3	pF
Data Output pin Capacitance	C_{DOUT}	-	0.1	pF

Timing Parameters [1] (T_J = -40 °C to 125 °C, V_{DD} = 1.62 V to 1.98 V, V_{SS} = 0V, C_{LOAD}=1pF)

User mode

Parameter	Symbol	SFD64KX32M64P4		Unit
		Min.	Max.	
X address access time	T_{xa}	-	35	ns
Y address access time	T_{ya}	-	35	ns
PROG/ERASE to NVSTR set up time	T_{nvs}	5	-	us
NVSTR hold time	T_{nvh}	5	-	us
NVSTR hold time(mass erase)	T_{nvh1}	100	-	us
NVSTR to program set up time	T_{pgs}	10	-	us
program hold time	T_{pgh}	20	-	ns
program time	T_{prog}	20	40	us
address/data set up time	T_{ads}	20	-	ns
address/data hold time	T_{adh}	20	-	ns
recovery time	T_{rcv}	1	-	us
cumulative program HV period[2]	T_{hv}	-	8	ms
Erase time	T_{erase}	20	40	ms
Mass erase time	T_{me}	20	40	ms

Notes:

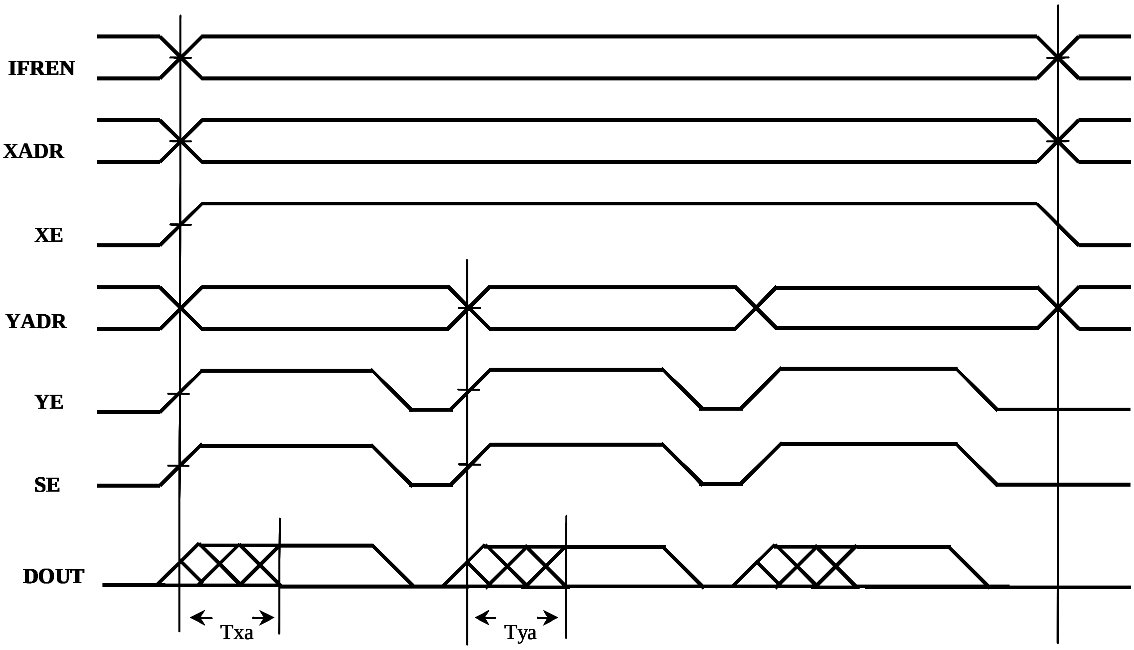
1. All timing numbers are pre-simulation and subject to change before silicon verification.
2. T_{hv} is the cumulative high voltage programming time to the same row before next erase and the same address can't be programmed more than twice before next erase.
3. All signals that align together in the timing diagrams should be derived from the same clock edge. Set up and hold times are not critical if they are within 10ns.
4. Reading operation is combinatorial. Sequences of timing edges are not important. The latest valid signal determines the access time.
5. All timing measurements are from the 50% of the input to 50% of the output.
6. All input waveforms have rising time(tr) and falling time(tf) of 1ns.
7. For capacitive loads greater than 1pF, access time will increase by 1ns per pF of additional loading.

Test mode

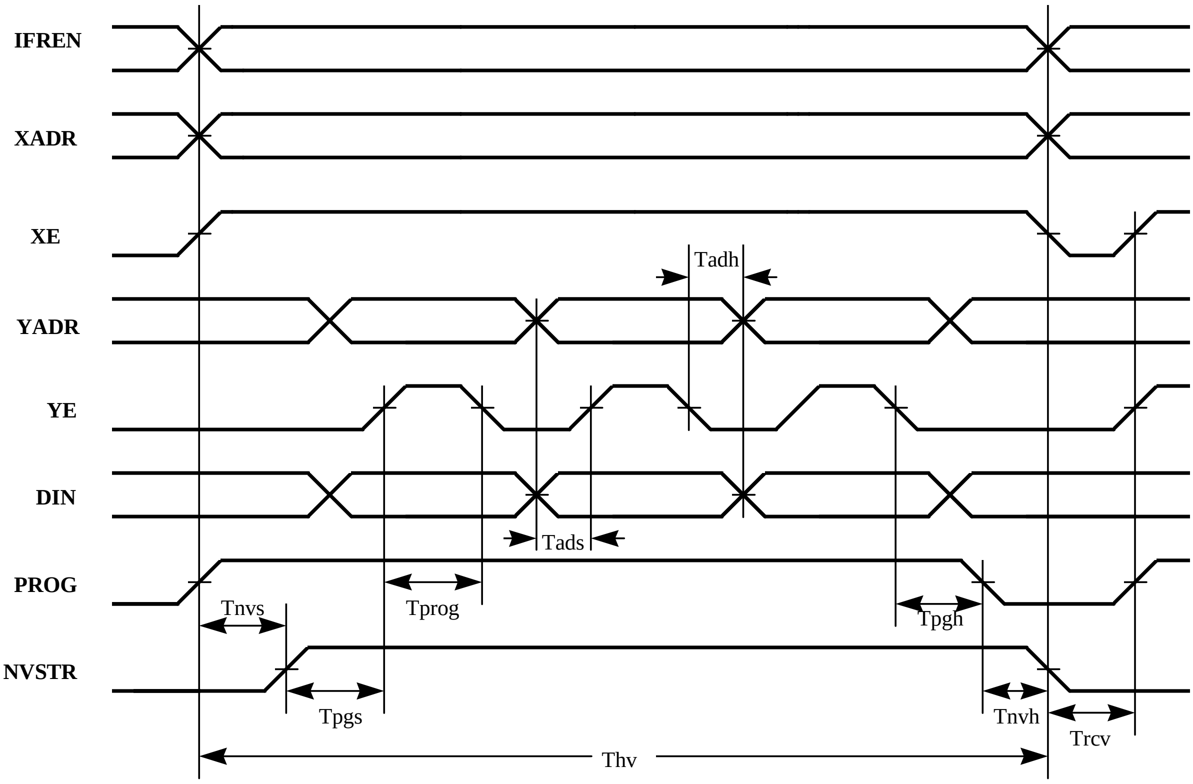
Parameter	Symbol	Value		Unit
		Min	Max	
TMR pulse width	Ttmr	20	-	ns
TMR to internal TMSE set up time	Trses	10	-	ns
Internal TMSE to data set up time	Tseds	10	-	ns
Data set up time for latch	Tlds	10	-	ns
Set latch pulse width	Tlpw	20	-	ns
Data hold time for latch	Tldh	10	-	ns
Data to internal TMSE hold time	Tdseh	10	-	ns

Timing Waveforms

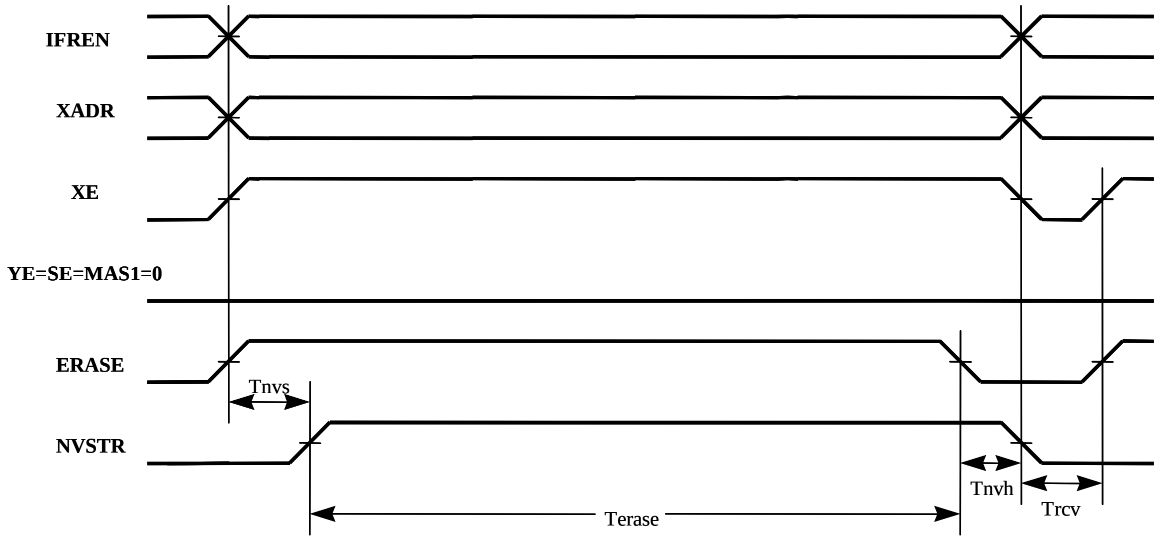
Read Cycle



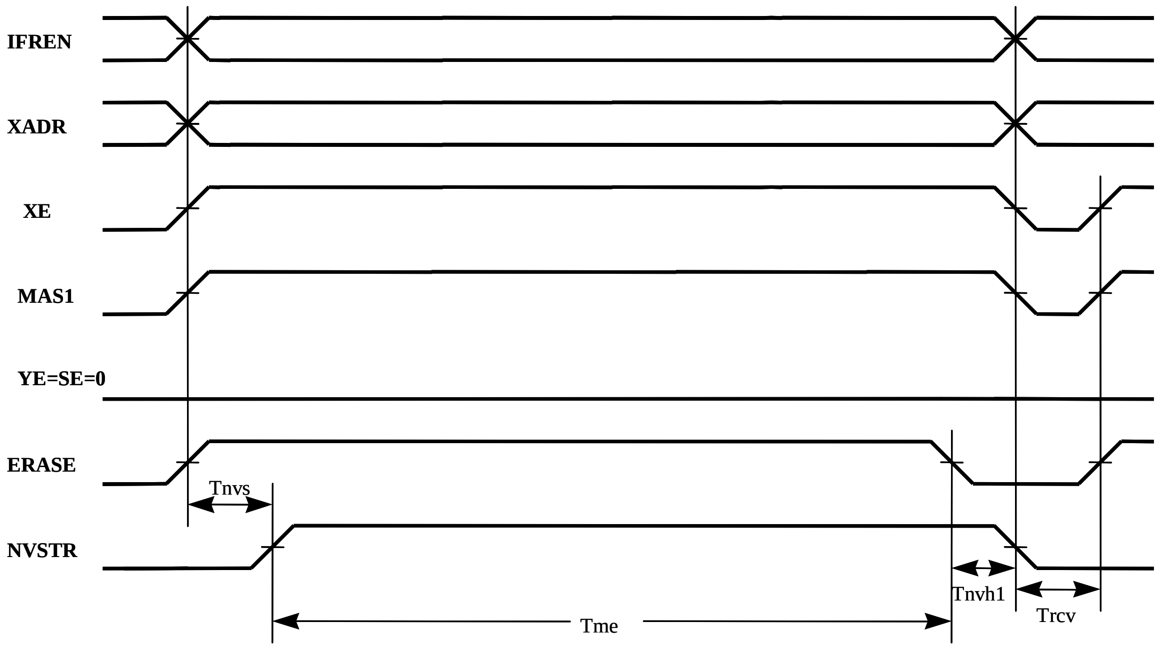
Program Cycle



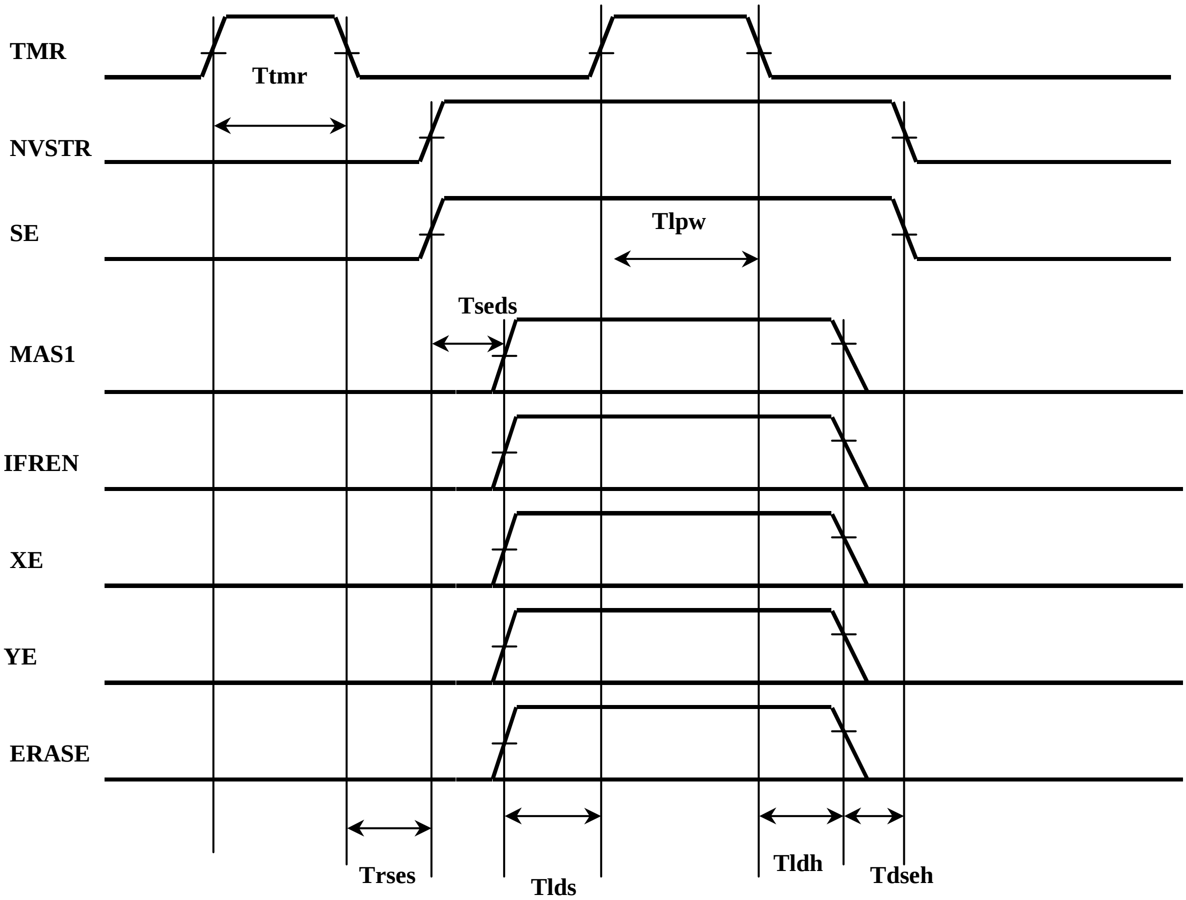
Erase Cycle



Mass Erase Cycle



Test mode



Update history:

Date	Rev	What	Who
02/15/2006	010A	New created.	SSLiu